DATA STRUCTURE & ALGORITHMS (CSIT124)

Introduction

by

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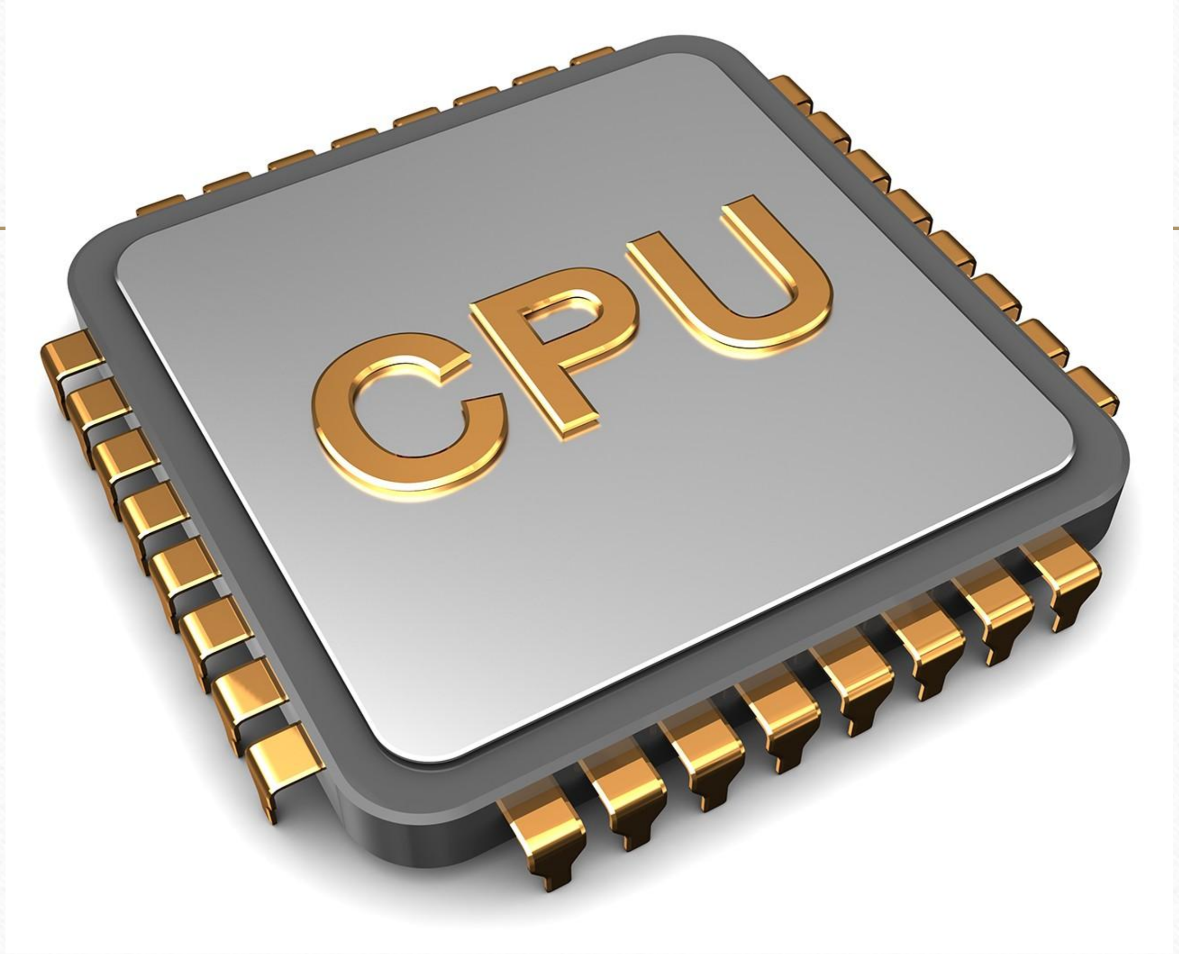
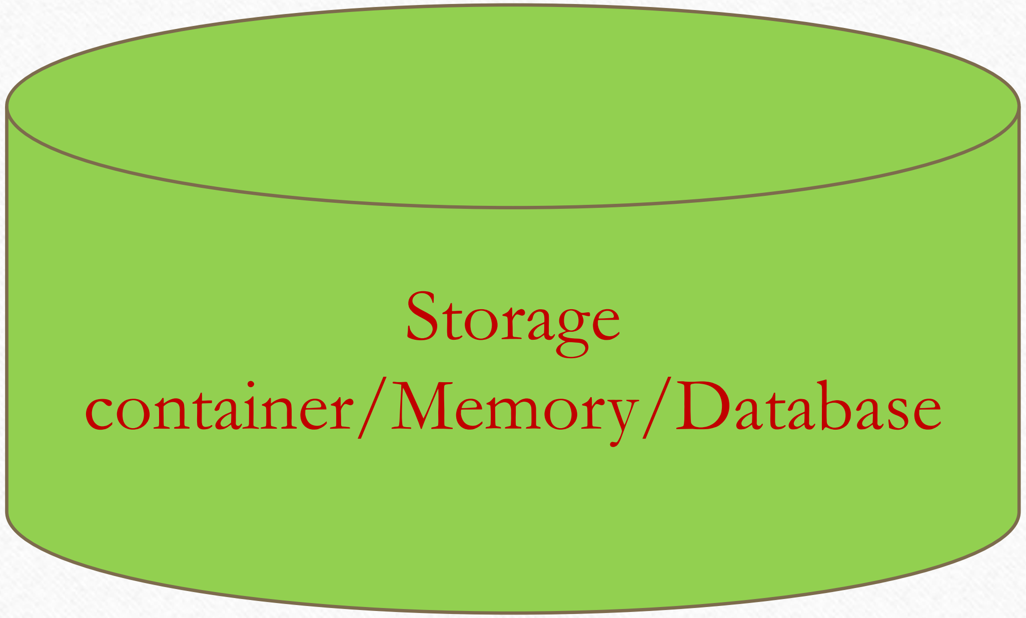
Computer?

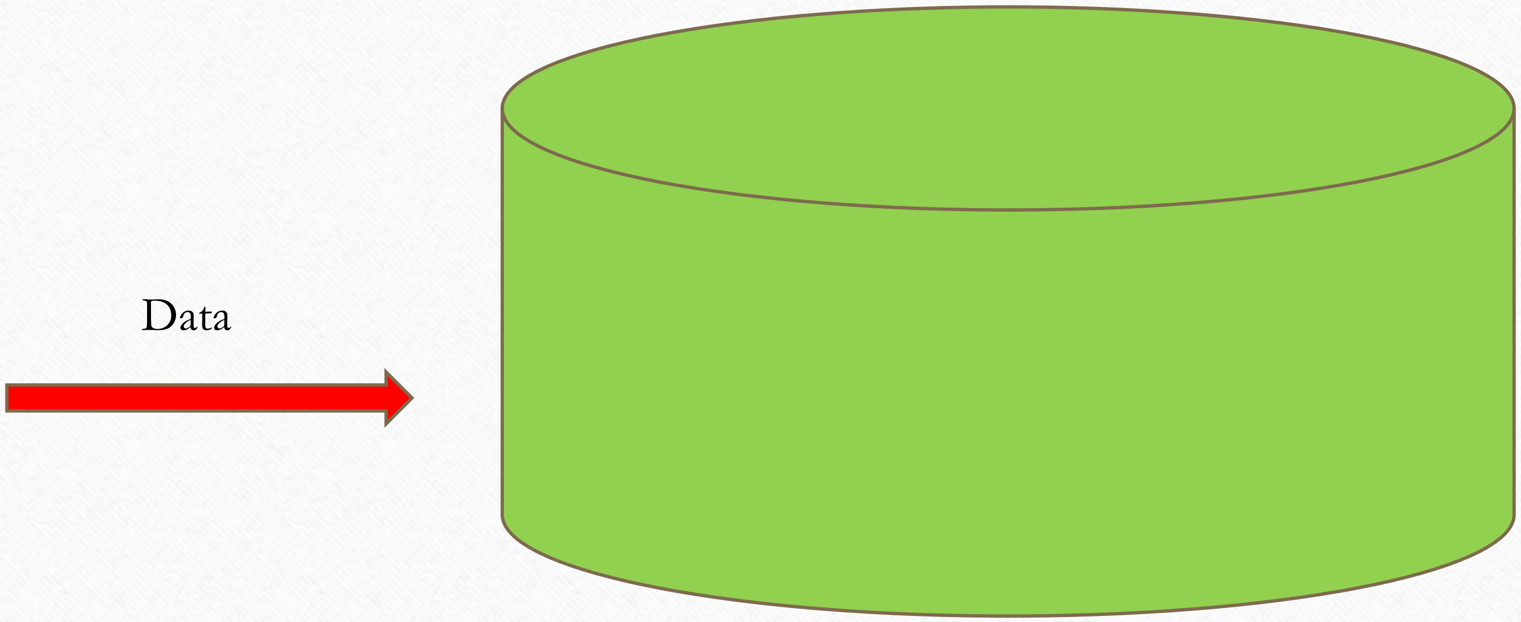
Input

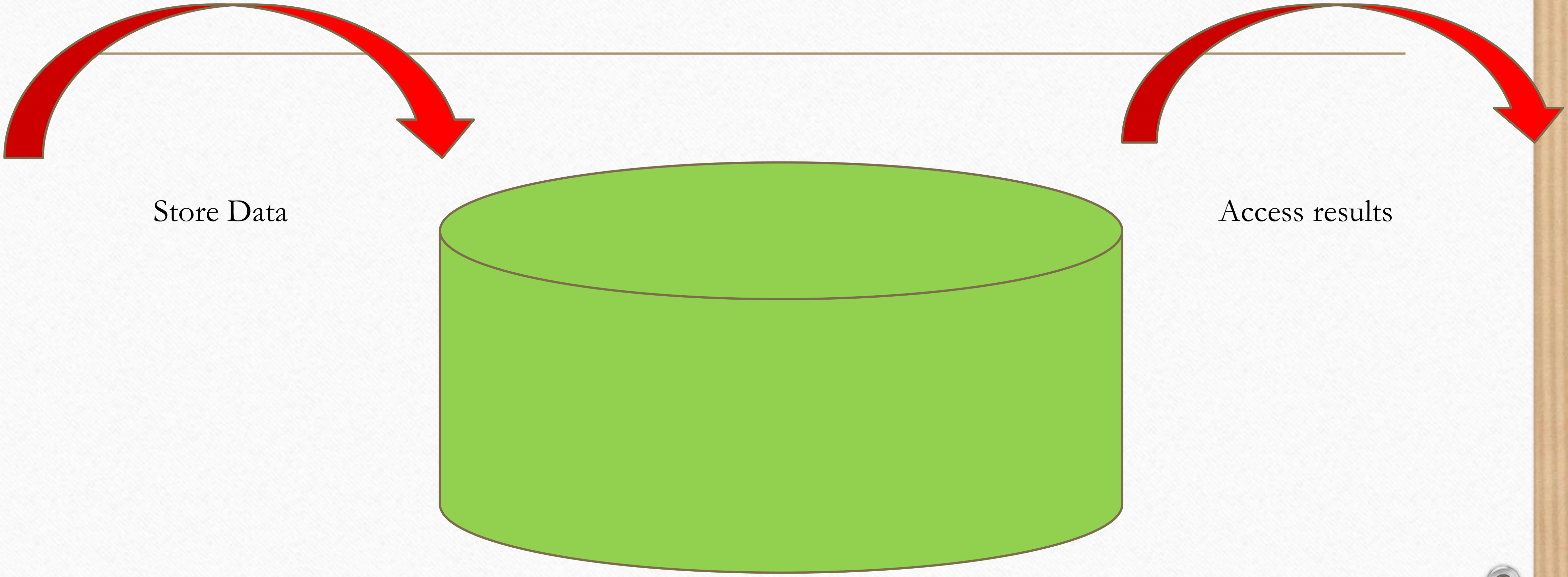
Computer?

Output


CPU

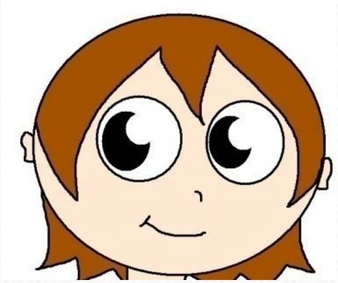

Storage
container/Memory/Database


Data


Store Data
Access results

What's data?

- Collection of facts/particulars
- Name: Asha, Nishit
- Email: abc@abc.com
- Phone: 1234567890



Data?

- Text
- Number
- Audio
- Video
- Speech
- Satellite images
- etc….

Information

- Well organized arrangement of data, not only a collection of data
- Name Ron my is.
- My name is Ron.

How does it look like?

Roll	Name	Year	Stream
1	ABC	2	CSE
2	XYZ	2	CSE
3	MNO	2	AI

What is Data Structure?

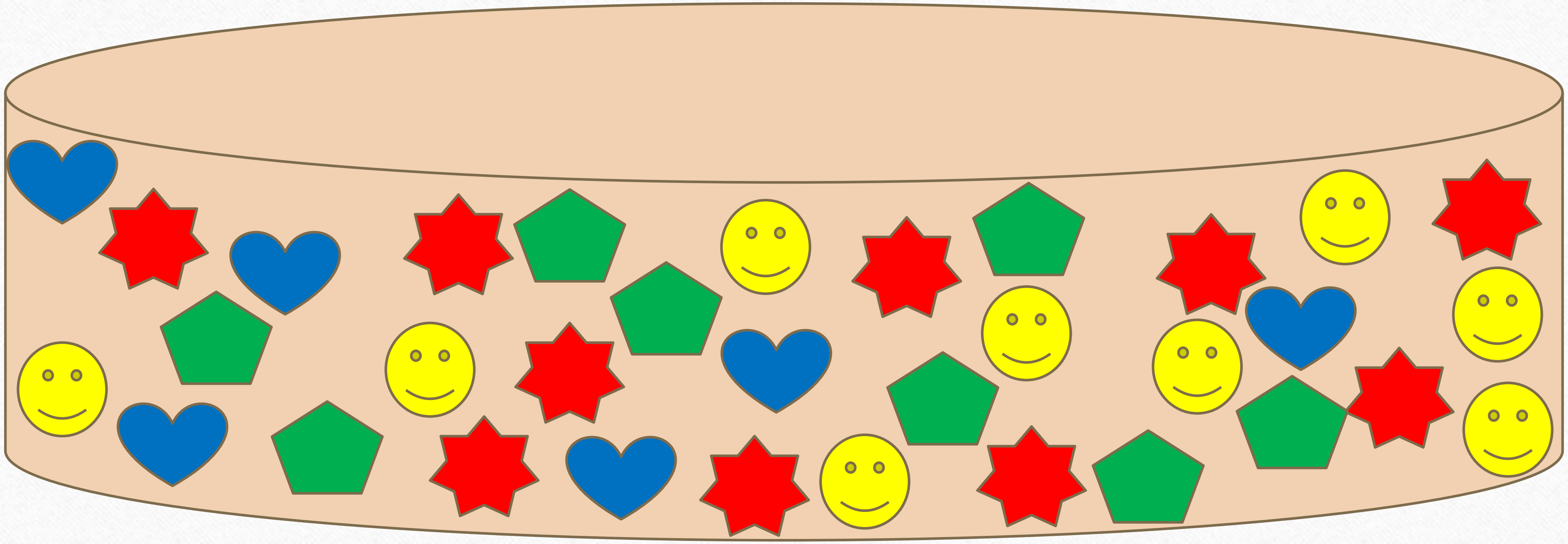
- A way of organizing data in such a way so that it can be easily accessible

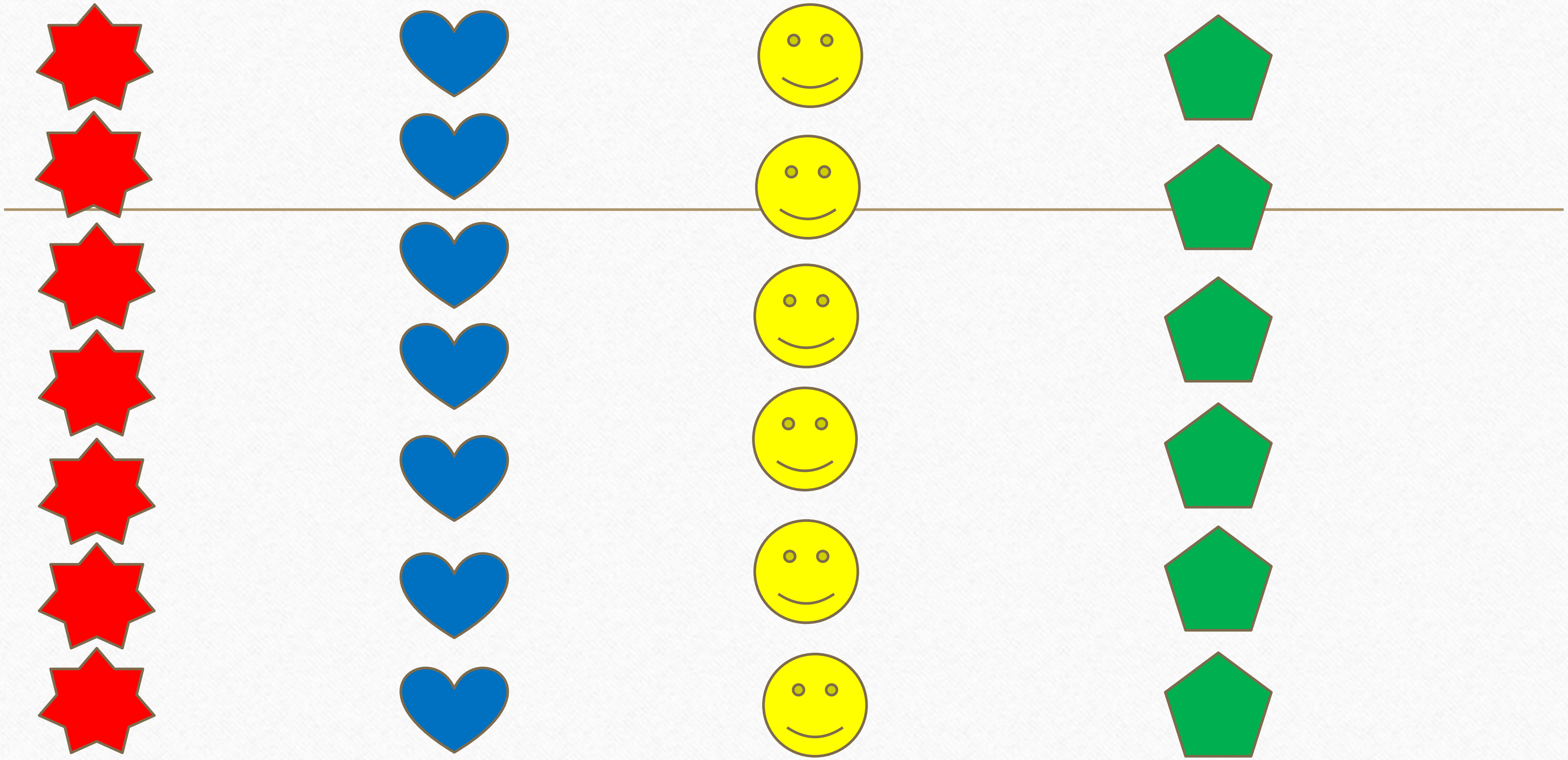
Why Data Structure?

Where is my blue shirt?

I like strawberry

How many red stars?

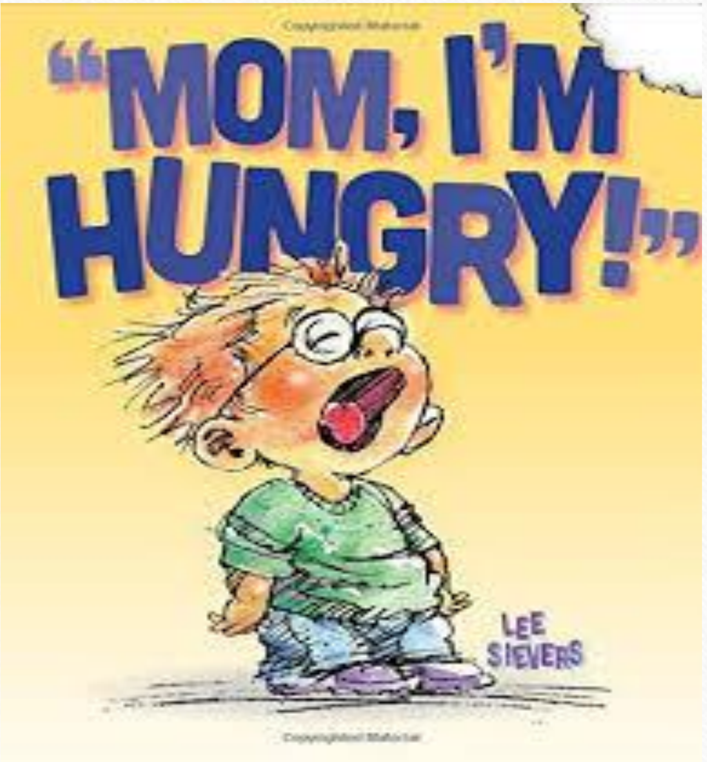




Solve complex problems

- **Problem:** Algorithm (step by step method) + Language (to talk to the computer)
- Time
- Space
- Role of Data Structure

Time matters



Space is also a concern

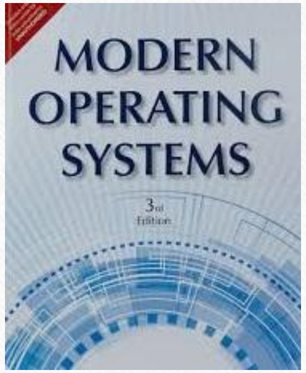
Data Structure?

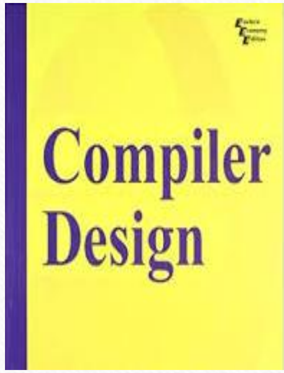
- Study of various ways of organizing data in a computer
- Data Structure = Organized data + Allowed Operations

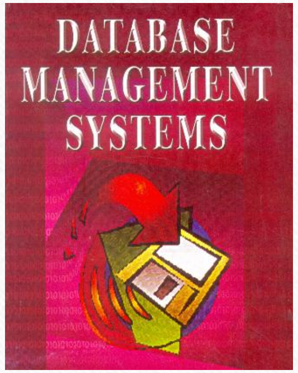
- Operations involve
- Adding
- Deleting
- Searching
- Sorting
- Merging
- Rearranging
- Traversing

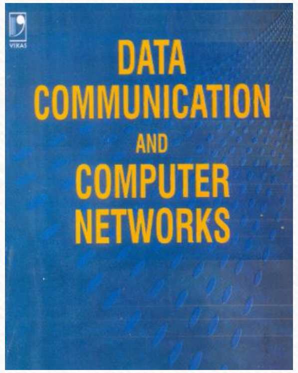
Importance of DS?

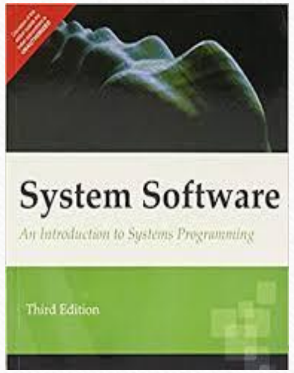
- Data Structures: An integral part of







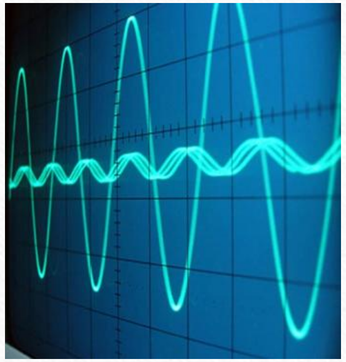




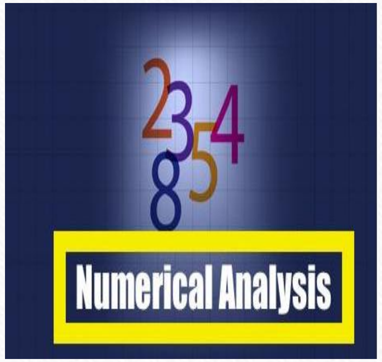
Application of DS?

- Data Structure+ Algorithm = Application

Image Processing

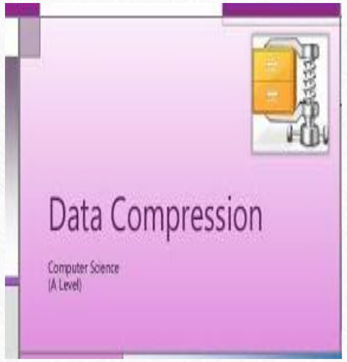


Digital Signal processing



Numerical Computations

Cryptography

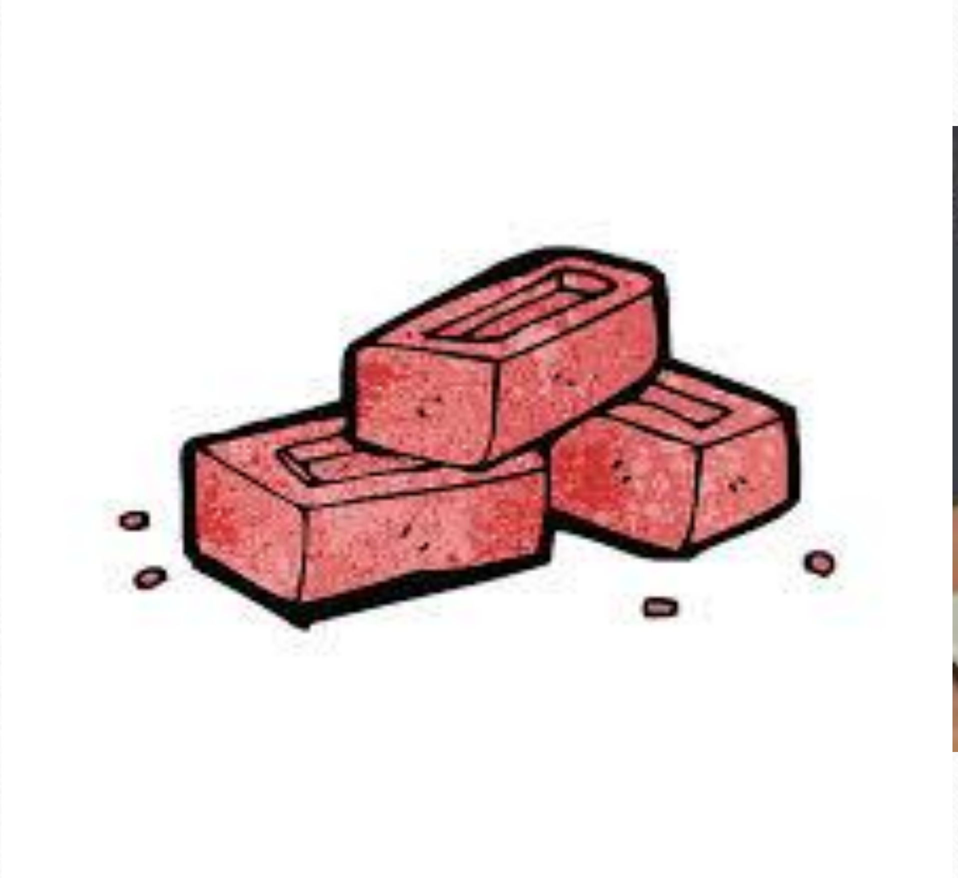


Data Compression

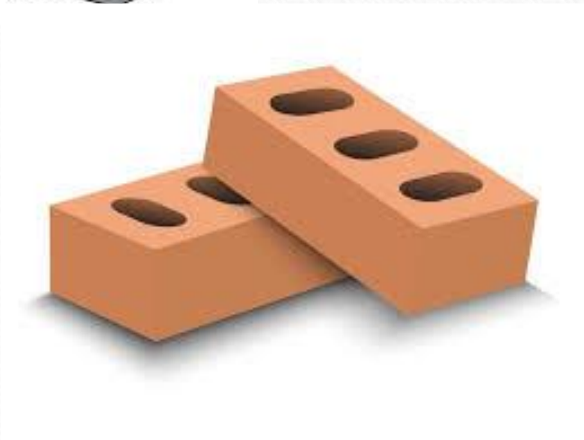
Genetic Studies

Classification of Data Structure

- Primitive Data Structure
- Non-Primitive Data Structure

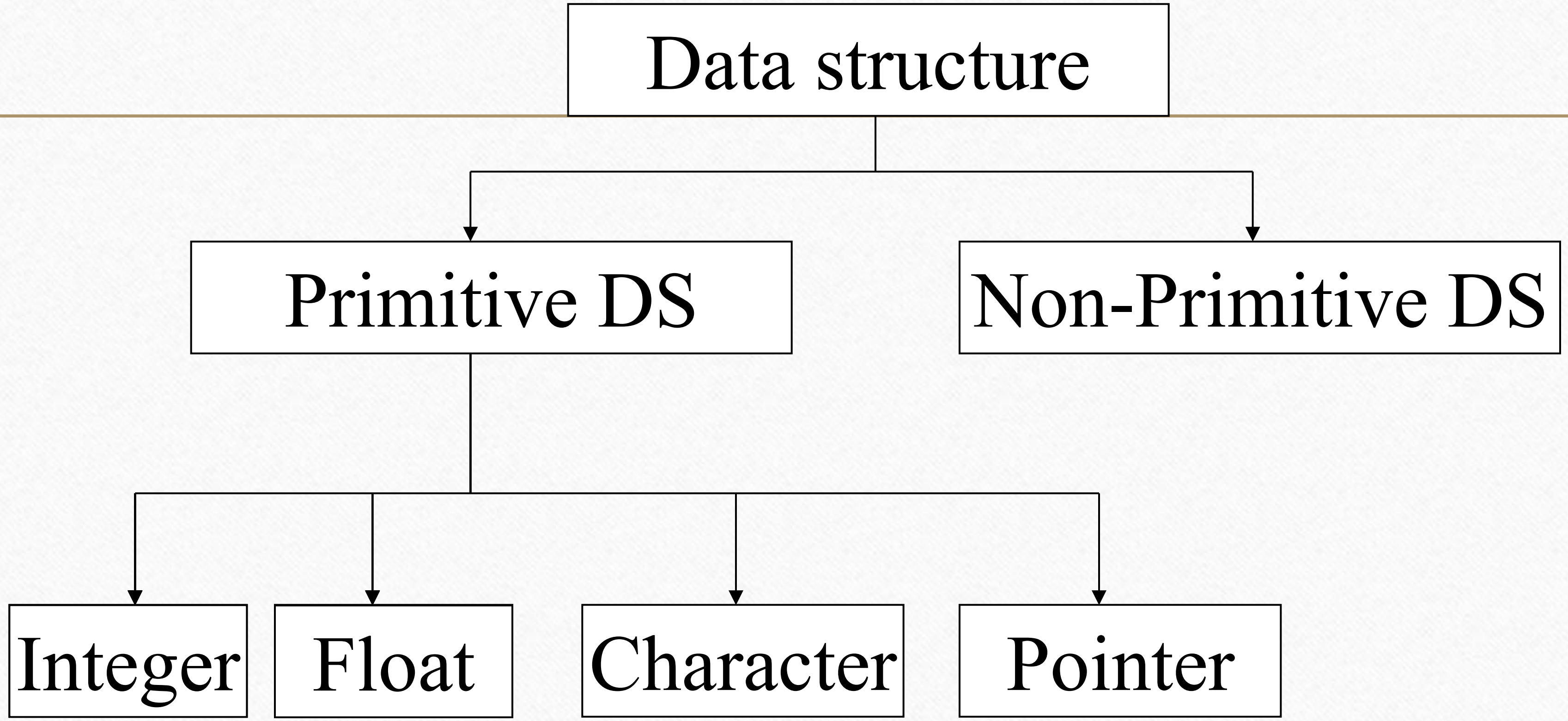


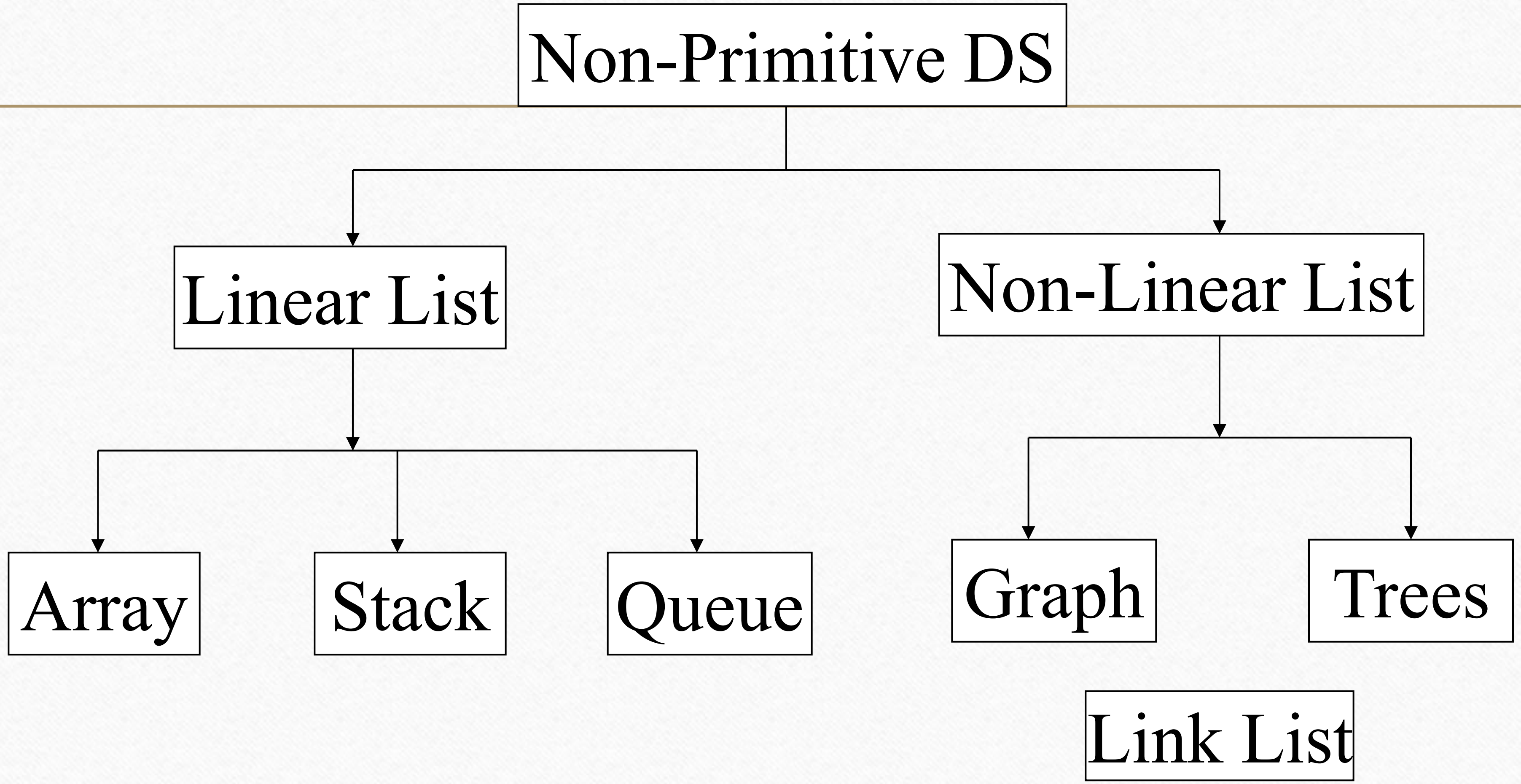

CEMENT




FACTORY

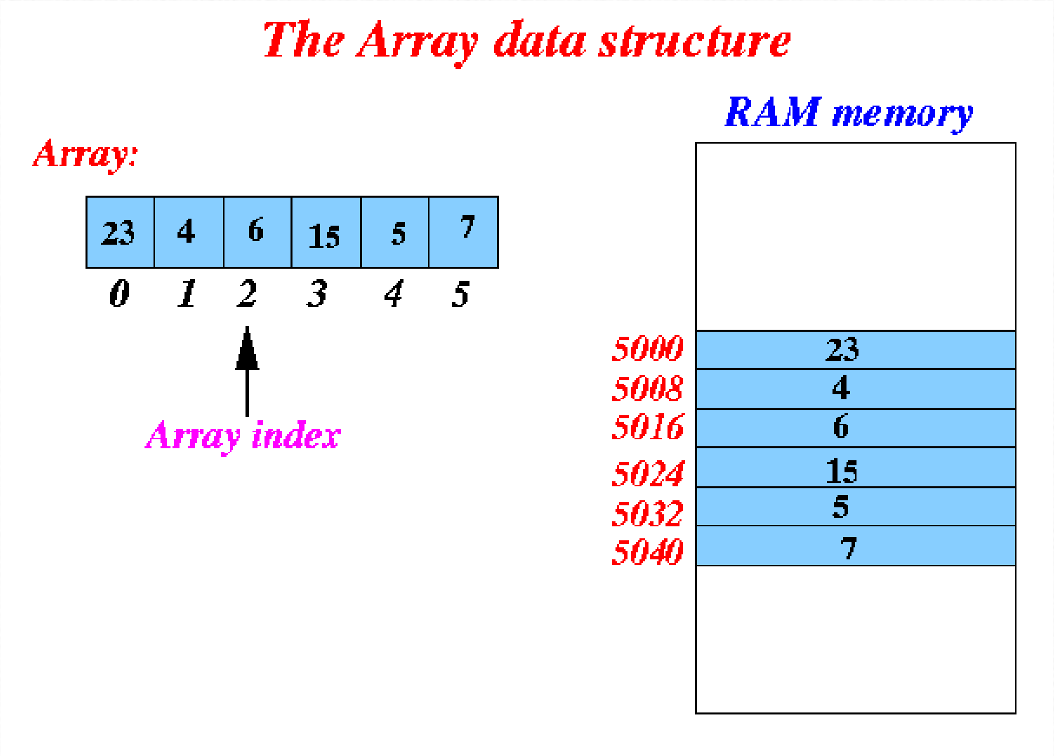
- A primitive data structure is generally a basic structure that is usually built into the language, Integer, Float, Char, Double
- A non-primitive data structure is built out of primitive data structures linked together in meaningful ways, Array, Stack, Queue, Linked-list, Tree and Graph


Data structure
Primitive DS
Non-Primitive DS
Integer
Float
Character
Pointer

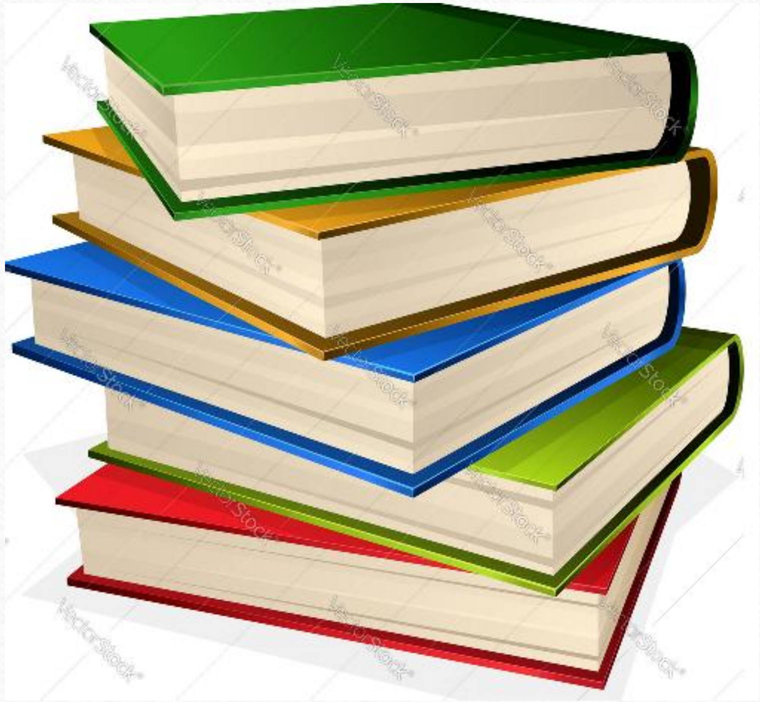

Non-Primitive DS
Linear List
Non-Linear List
Array
Stack
Queue
Graph
Trees
Link List

Array

Array

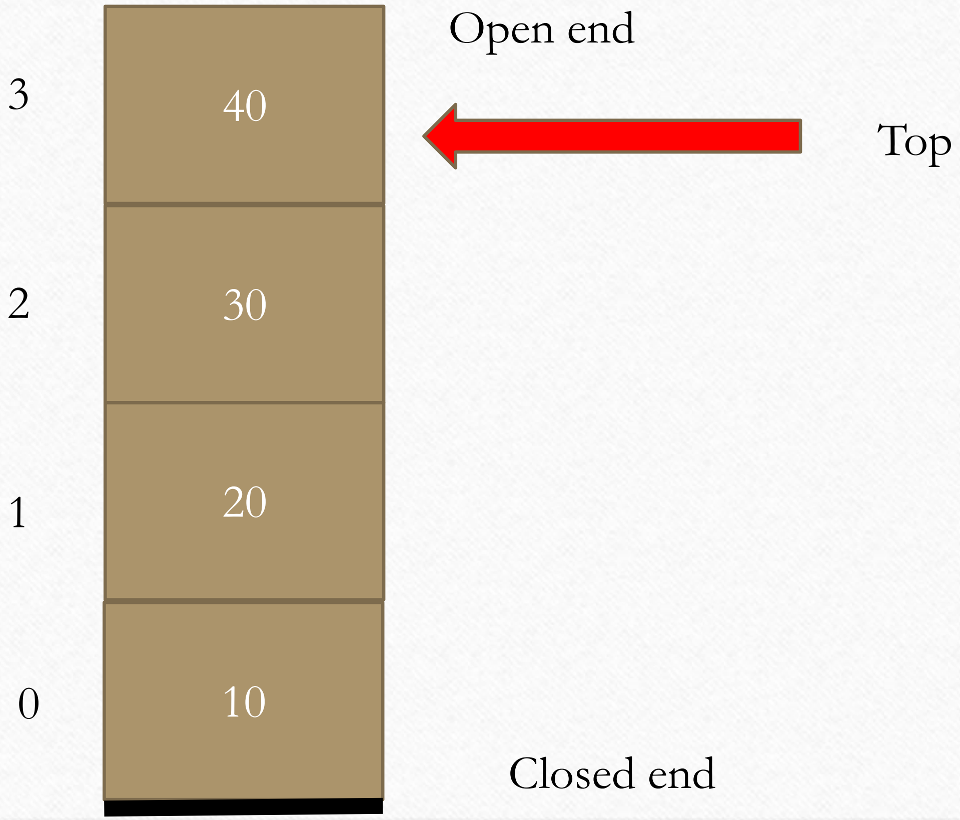

The Array data structure
Array:
23
4
6
15
5
7
0
1
2
3
4
5
Array index
RAM memory
5000
5008
5016
5024
5032
5040
23
4
6
15
5
7

Stack



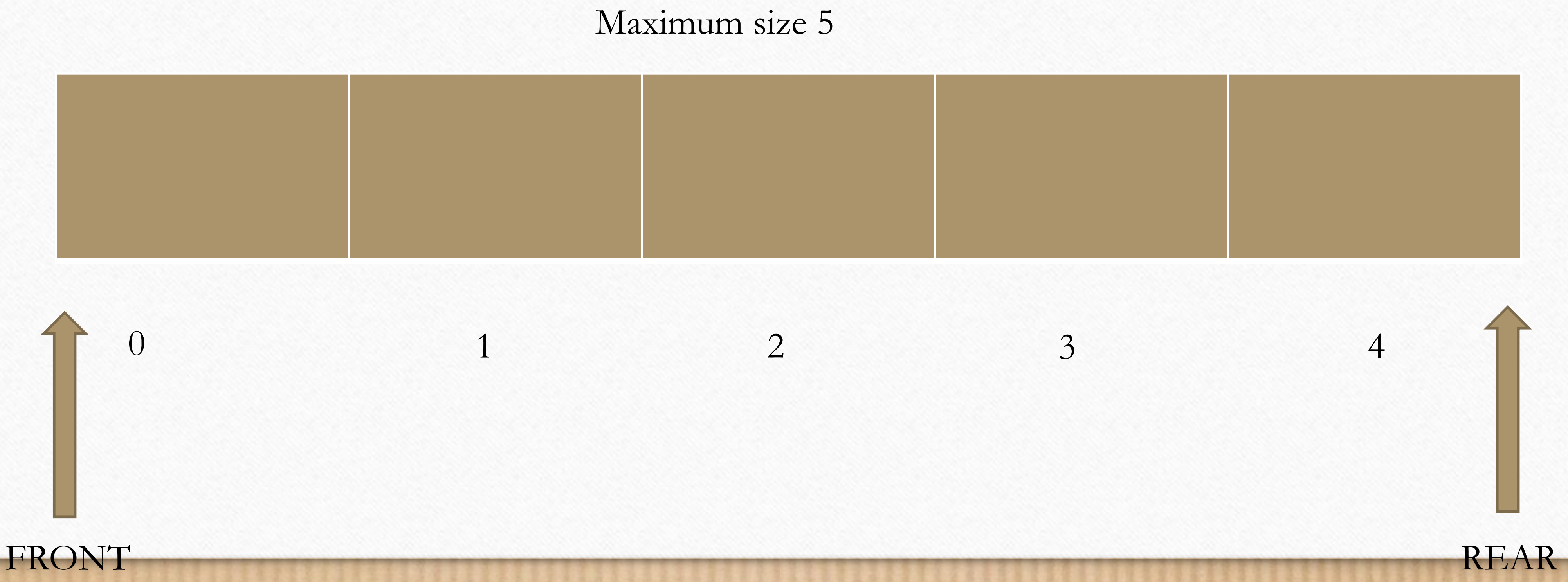
Stack

Fixed size

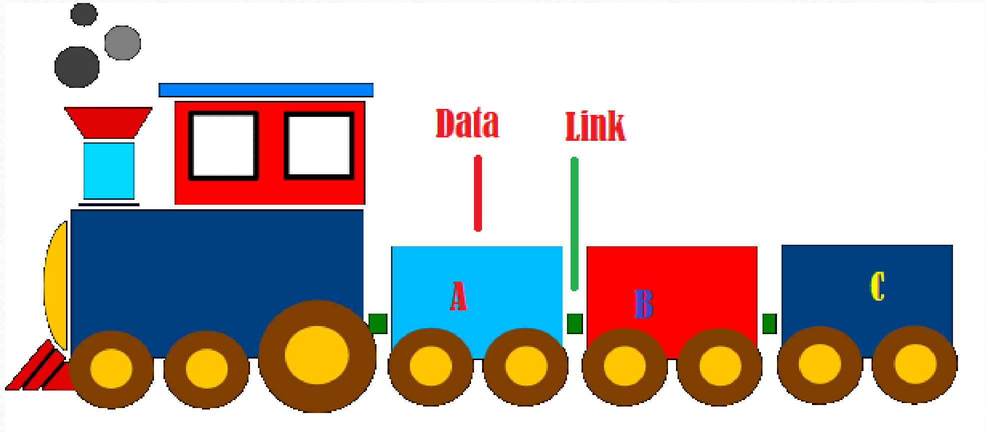


Queue

Simple Queue by Array representation

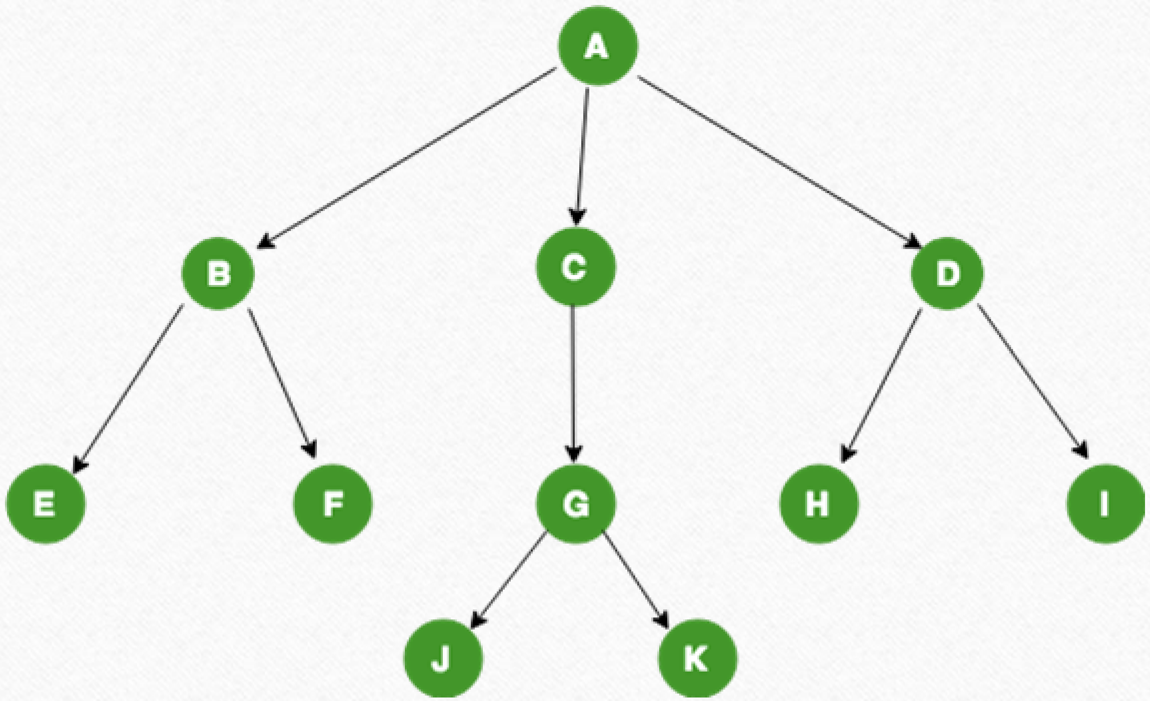


Linked List



Tree

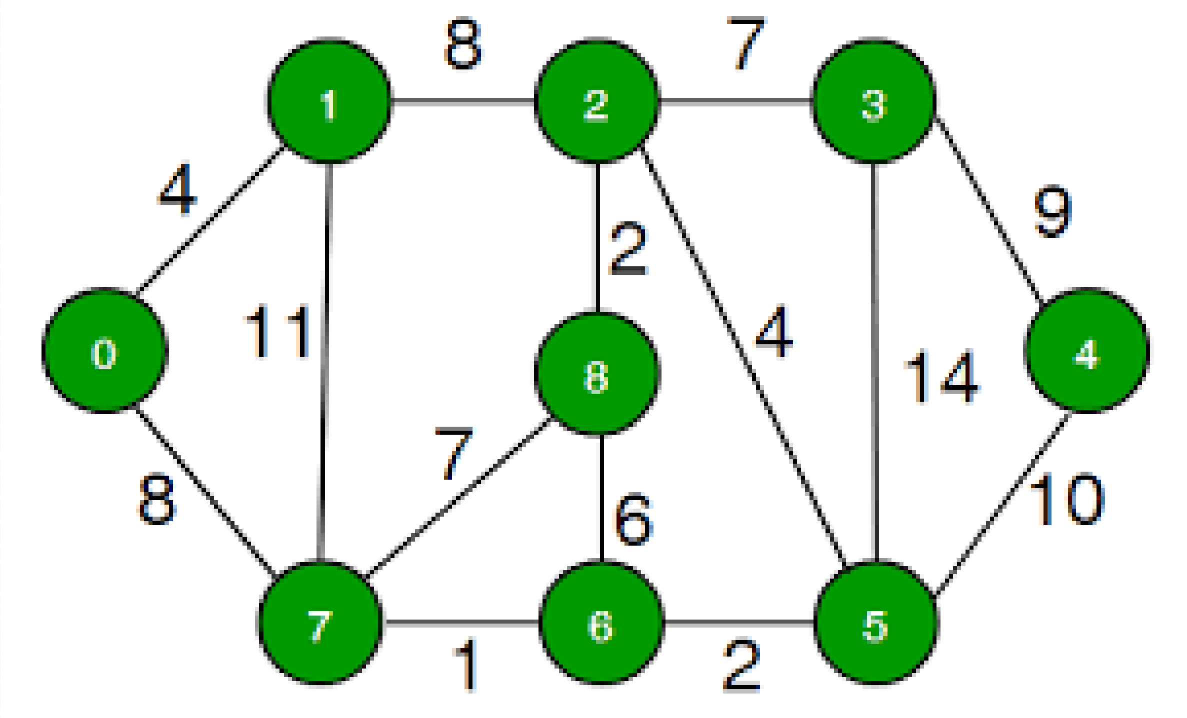
Tree



Graph


1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
Niayesh Hwy
Darya Blvd
Modiriyat Blvd
Parsian Hospital

Graph



Next Lecture

- Stack

Thank you